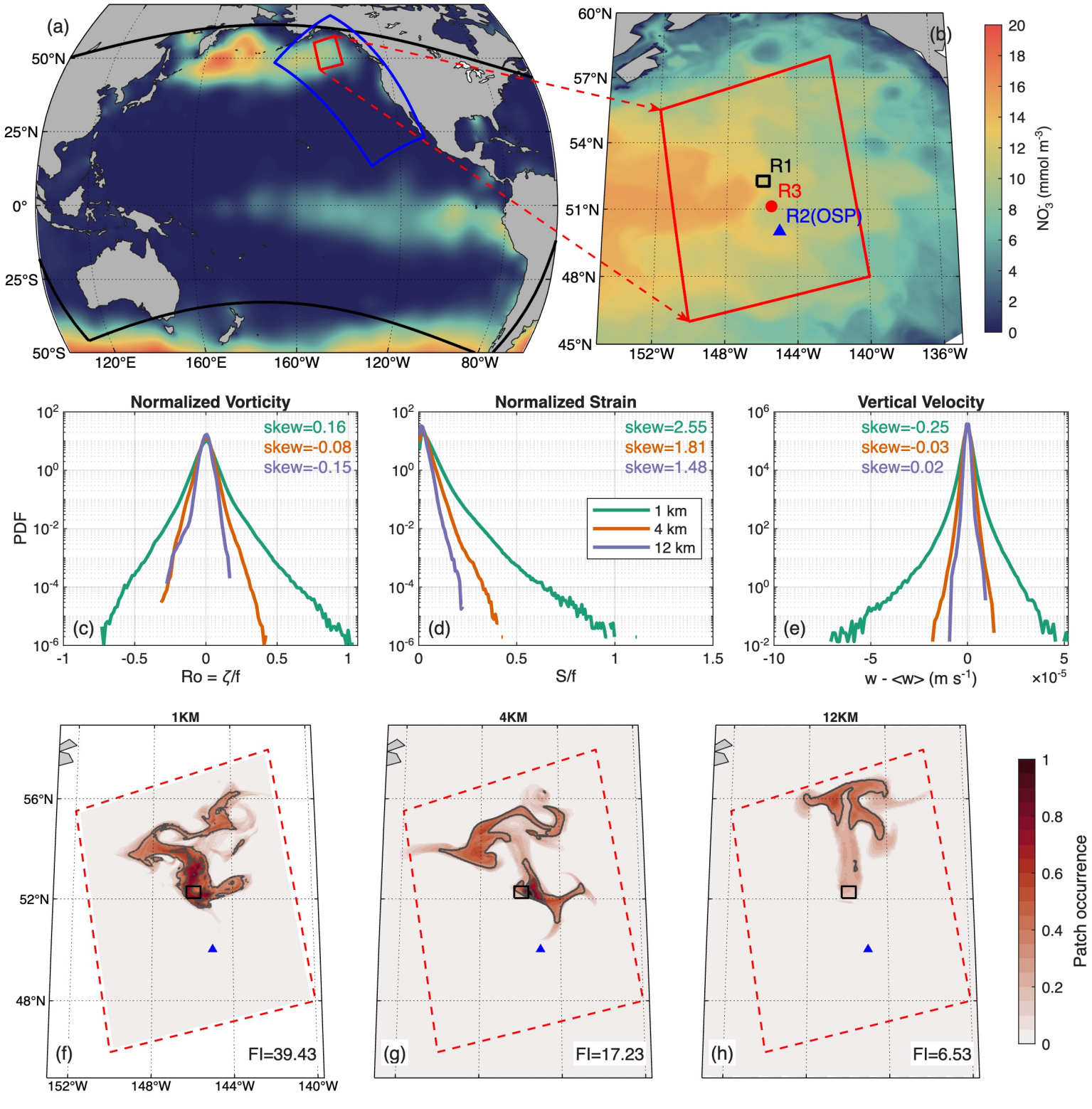

(a)
50°N
25°N
0°
25°S
50°S
120°E
160°E
160°W
120°W
80°W
(b)
60°N
57°N
54°N
51°N
48°N
45°N
152°W
148°W
144°W
140°W
136°W
R1
R3
R2(OSP)
NO3 (mmol m-3)
20
18
16
14
12
10
8
6
4
2
0
Normalized Vorticity
skew=0.16
skew=-0.08
skew=-0.15
PDF
(c)
Ro = ζ/f
Normalized Strain
skew=2.55
skew=1.81
skew=1.48
1 km
4 km
12 km
(d)
S/f
Vertical Velocity
skew=-0.25
skew=-0.03
skew=0.02
(e)
w - <w> (m s-1)
×10-5
1KM
4KM
12KM
56°N
52°N
48°N
152°W
148°W
144°W
140°W
(f)
FI=39.43
(g)
FI=17.23
(h)
FI=6.53
Patch occurrence
1
0.8
0.6
0.4
0.2
0